

Lab 3: Architectural Style Justification

Architecture Selection

For our Restaurant Table Booking & Pre-Ordering App, we picked a layered architecture. We split the system into five components, each with its own job: the Diner App/UI, the Reservation Manager, the Pre-Order Service, the Table & Floor Plan Manager, and the Restaurant Database.

Reason 1: Separation of Responsibilities

The app does a few different things, so we didn't want them all mixed together. The Diner App only deals with what the user sees and clicks. The Reservation Manager handles booking tables, the Pre-Order Service takes care of food orders and how long they take to prepare, and the Table & Floor Plan Manager keeps track of which tables are free. Keeping these apart makes the system easier to build and test. It is also easier to change one part later without breaking the others.

Reason 2: Consistent Reservation and Table Data

A booking system needs its data to be correct. If two people can book the same table at the same time, the app stops being useful. With a layered design, the reservation and table components all go through set interfaces to the same central database. Because everyone reads from one place, table statuses stay consistent and double bookings are much less likely.

Security Advantage

In our design the Diner App never talks to the Restaurant Database directly. Diners only use the reservation and pre-order services, and those services are the ones that touch the data. This means we can put access control in the service layer. For example, a normal diner wouldn't be allowed to change a table's status or see the restaurant's management data.

Performance Benefit

The Table & Floor Plan Manager gives the Diner App the current table availability through its own interface, so the app doesn't have to query the database every time the user does something. Things that are requested a lot, like the menu and table info, can also be handled in the application layer. This should keep table selection and reservations feeling quick for the user.